

Project 2: Building a Collaborative Multi-Agent System with LangGraph

Instructor

Dipanjan Sarkar

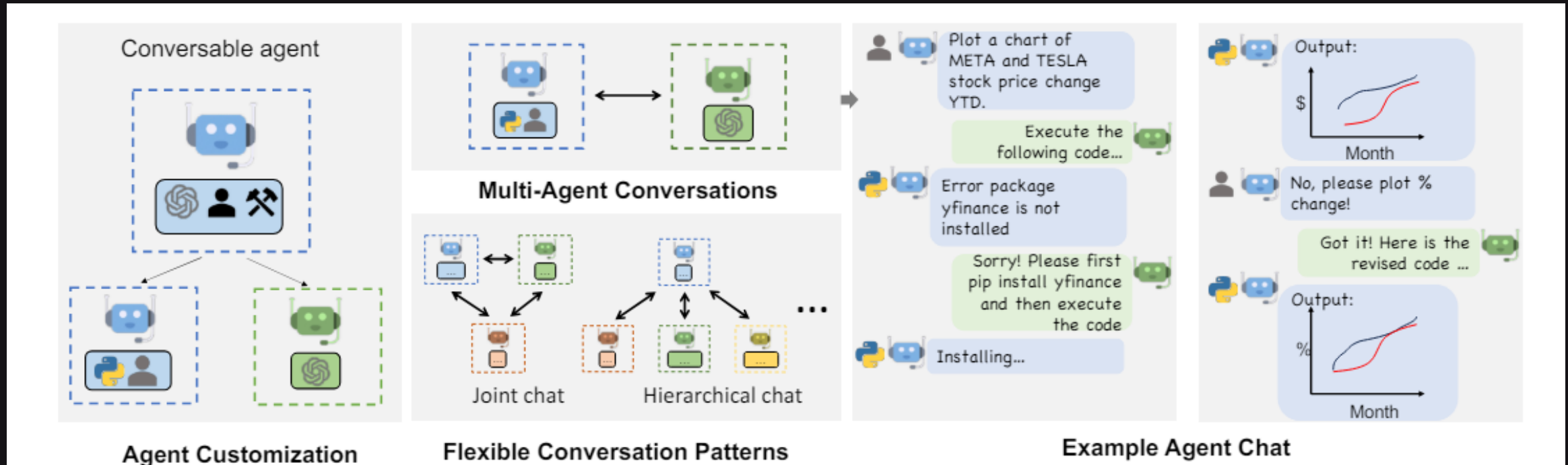
Head of Community & Principal AI Scientist at Analytics Vidhya

Google Developer Expert - ML & Cloud Champion Innovator

Published Author

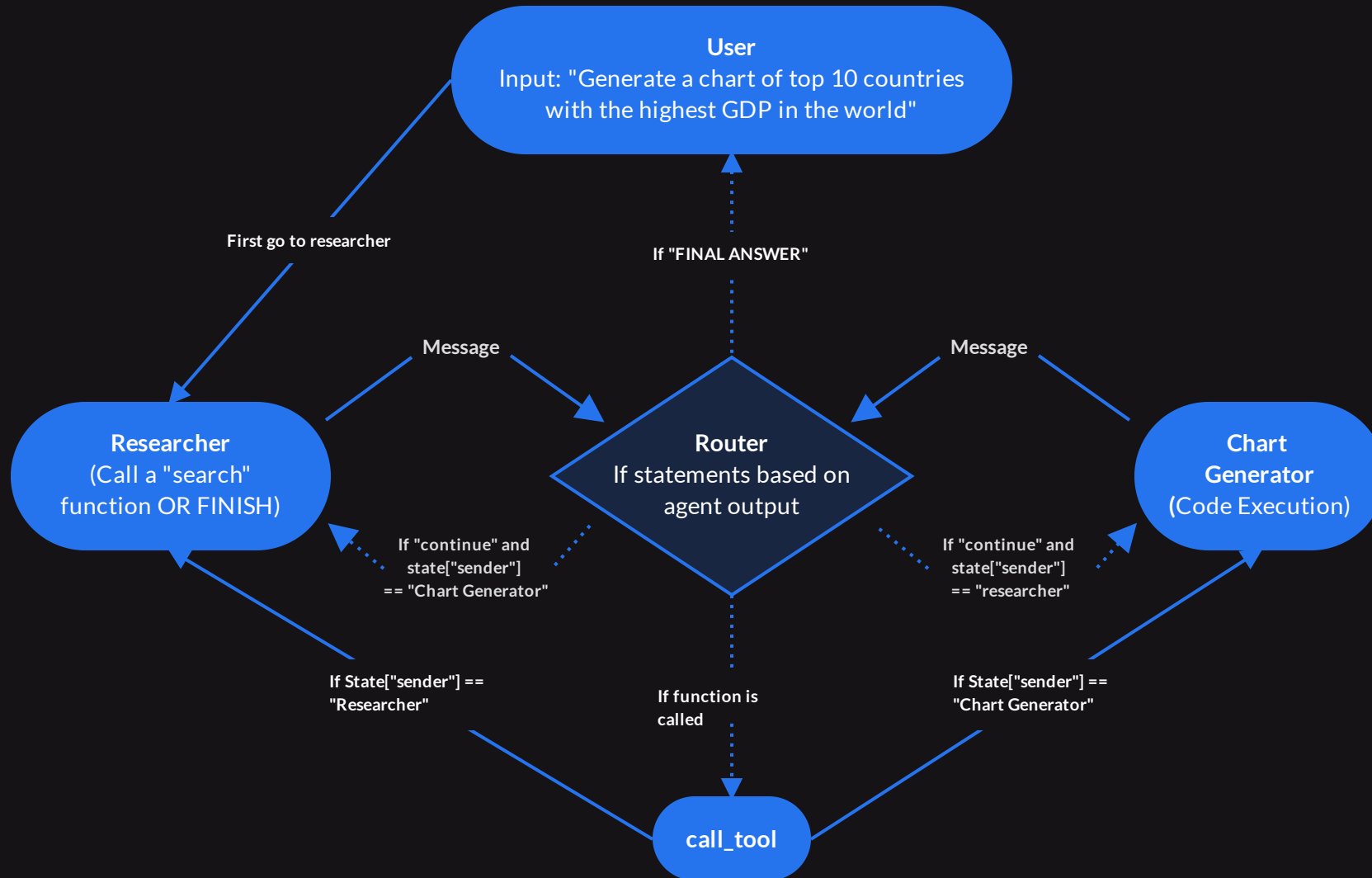


What are Multi-Agent Systems?



- Multi-Agent Systems are systems in which multiple agents collaborate or work independently to solve specific tasks.
- **Microsoft's Paper:** "AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation" is a framework designed to facilitate these interactions seamlessly.

Collaborative Multi-Agent workflow to implement



- **Step 1 - User Query**

- User gives a query to the multi-agent system

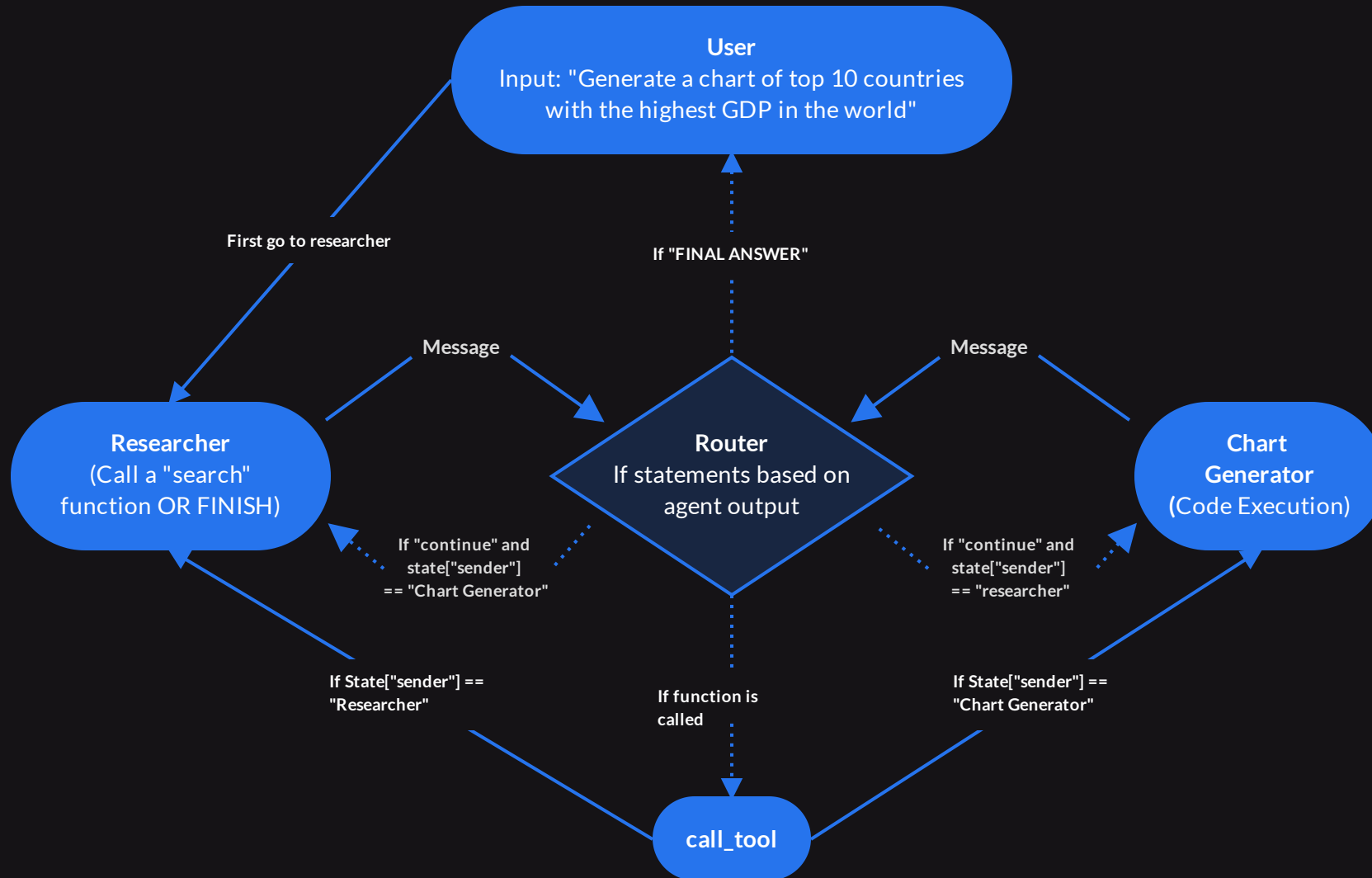
- **Step 2 - Researcher Agent Node**

- Uses web search tool to get information from the web for user query

- **Step 3 - Chart Generator Agent Node**

- Uses python code and takes data from researcher node to build data visualization

Collaborative Multi-Agent workflow to implement



- **Step 4 -Routing Node**
 - If some information is missing for visualization go back to researcher node to get more data
- **Step 5 - Routing Node**
 - If any new information is obtained from researcher go to the chart generator node to recreate visualization
- **Step 6- END Node**
 - If visualization is successfully created, end the agent run